

WHERE DECODER COSINE SIMILARITY FAILS FOR SAE FEATURE FLOW DISCOVERY

Hendrik Droste¹, Christian Medeiros Adriano¹, Kathrin Korte², Holger Giese¹

¹Hasso Plattner Institute, Germany, hendrik.droste@uni-potsdam.de

²IT University of Copenhagen, Denmark, kort@itu.dk

ABSTRACT

Foundation models are increasingly adapted through fine-tuning, model editing, and alignment procedures while retaining previously acquired capabilities. Understanding the internal computations that support these adaptations is therefore becoming increasingly important for continual model evolution. Sparse autoencoders (SAEs) provide interpretable feature dictionaries for residual-stream activations and sublayer outputs, but it remains unclear how state features and update features interact to produce downstream residual features. In this work, we focus on MLP updates as a first test case. We construct a transition atlas of triples $s_k + u_j \rightarrow t_\ell$, where a residual-state feature and an MLP-update feature jointly predict a target residual feature, and validate candidate triples by ablating the decoded update feature. In a 20M-token Pythia-160M $L_7 \rightarrow L_8$ run, we find 38,125 strong ablation-effect transitions, but 88.0% have both state-target and update-target decoder cosine similarity below 0.7. As a preliminary cross-model check, a run of 20M-token Gemma-3-4B $L_{21} \rightarrow L_{22}$ causally validates only the top 30,000 ranked candidate triples by ablating the decoded update feature, and 53.6% of strong-effect triples have both state-target and update-target decoder cosine similarity below 0.7. The Gemma result is directionally consistent with Pythia, but weaker, since update-target cosine recovers many of the strongest Gemma effects and the run is not a full-atlas causal validation. Ultimately, our results suggest that feature flow atlases can serve as diagnostics of representation-update mechanisms and thereby inform tools for steering model updates. Future work will validate more complex patterns across layers, models, and SAE families.

1 INTRODUCTION

Large foundation models are increasingly adapted after pretraining through fine-tuning, parameter-efficient methods, model editing, and incremental knowledge integration, rather than deployed as fixed systems (Zheng et al., 2025; Zhao et al., 2026). These settings require incorporating new information while preserving prior capabilities, echoing the stability–plasticity trade-off in continual learning (Kirkpatrick et al., 2017; Wickramasinghe et al., 2023). Existing work can localize factual associations in feed-forward modules (Meng et al., 2023) and identify concepts that shift or emerge during chat tuning (Minder et al., 2025), but it offers less insight into the causal feature-level transformations by which internal computations are reused, refined, or suppressed.

Sparse autoencoders provide interpretable decompositions of transformer representations into sparse semantic features (Cunningham et al., 2023), and recent work studies how such features persist, transform, or emerge across layers (Balcells et al., 2024; Laptev et al., 2025). We therefore treat feature flow discovery as a mechanistic step toward understanding how residual-state features and MLP-update features interact during computation.

We construct a transition atlas of triples $s_k + u_j \rightarrow t_\ell$, where a residual-state feature s_k and an MLP-update feature u_j jointly predict a downstream residual feature t_ℓ . With this atlas we answer the following **research question**: Are strong cross-layer feature transitions recoverable from pairwise decoder geometry alone, or do causal transitions depend on the joint presence of a prior residual-state feature and an MLP-update feature?

2 FEATURE FLOW DISCOVERY

Our feature flow atlas contains transitions in the form of $s_k + u_j \rightarrow t_\ell$, where s_k is the source residual state feature, u_j is the target-layer MLP update feature, and t_ℓ is the target residual feature. We use decoder cosine similarity as a pairwise geometry baseline for the atlas. For each atlas triple, we compute the state-target decoder cosine $c_{s,t}$ and the update-target decoder cosine $c_{u,t}$. Following Laptev et al. (2025), we identify the most cosine-similar state and update

features for each target feature. Together, these comparisons ask whether the pairwise decoder cosine alone would recover the same transitions found by the atlas.

2.1 MATCHING PIPELINE

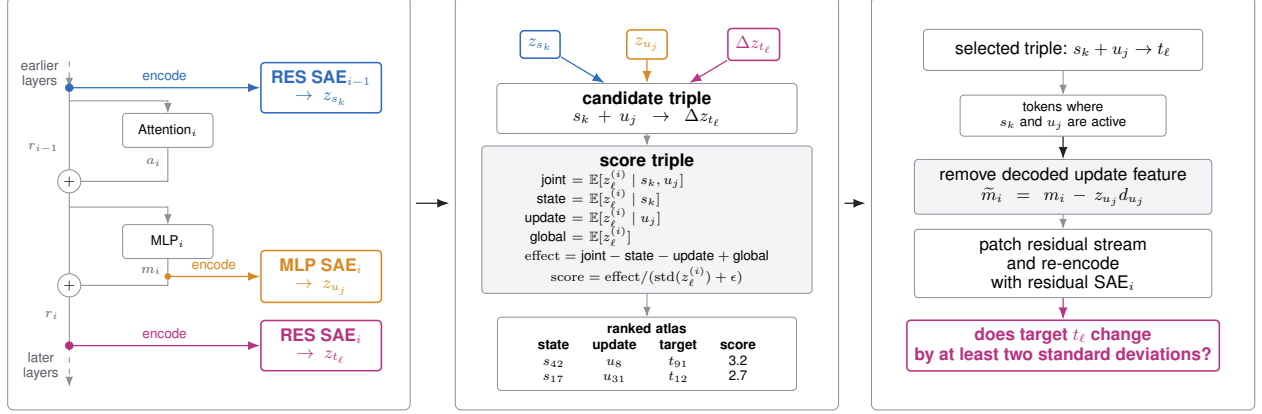


Figure 1: Overview of the feature matching pipeline. SAEs trained on the residual stream before layer i , the MLP output of layer i , and the residual stream after layer i define three separate feature spaces: state features s_k , update features u_j , and target features t_ℓ , where k, j , and ℓ index features within their respective SAE dictionaries. Activation frequencies over tokens are used to score triples $s_k + u_j \rightarrow t_\ell$, measuring whether update feature u_j predicts a target-feature change beyond what is expected from state feature s_k alone. High-scoring triples are then validated by ablating the decoded update feature and checking whether the target feature changes by at least two target-feature standard deviations.

We created the atlas using a three-stage feature matching pipeline. It requires SAEs trained for the state, update and target features. In our case we trained TopK SAEs on the residual stream after layers 7 and 8 as well as the MLP output of layer 8 of the Pythia-160M model (Biderman et al., 2023). To train stable SAEs, we adapted the training configuration from Song et al. (2025). The hyperparameters are listed in Appendix Section A.3.

Stage one of the pipeline is sparse event extraction. We run the model and collect the source residual stream, MLP-sublayer, target residual stream activations, and encode these with the corresponding SAEs. To reduce runtime we only store the top 64 active features per token.

The second stage constructs a feature-update candidate atlas by reading the sparse activations and turning them into ranked transition triples. We estimate, for each layer- i residual SAE feature, how much the layer- i MLP changes that feature. To do this, we encode both the final residual stream and the pre-MLP residual stream in the same layer- i residual SAE basis:

$$z_{\text{after}}^{(i)} = E_{\text{res}}^{(i)}(r_{\text{post}}^{(i)}), \quad z_{\text{no-mlp}}^{(i)} = E_{\text{res}}^{(i)}(r_{\text{post}}^{(i)} - m^{(i)}),$$

and define the MLP-induced target-feature change as

$$\Delta z^{(i, \text{mlp})} = z_{\text{after}}^{(i)} - z_{\text{no-mlp}}^{(i)}.$$

If the MLP sublayer does not change target feature t_ℓ , then $\Delta z_{t_\ell}^{(i, \text{mlp})} = 0$.

Next, we accumulate pairwise baseline frequencies for state-target and update-target relationships. These baselines prevent the atlas from ranking triples only because their individual features are common. Only after measuring them can we ask whether the full triple (s_k, u_j, t_ℓ) exists in which all the features could be causally related.

After accumulating the pairwise baselines and joint triple frequencies, we convert each candidate triple into a normalized score. For this, we compute

$$\text{interaction_score}(s_k, u_j, t_\ell) = \frac{\mathbb{E}[z_{t_\ell}^{(i)} | s_k, u_j] - \mathbb{E}[z_{t_\ell}^{(i)} | s_k] - \mathbb{E}[z_{t_\ell}^{(i)} | u_j] + \mathbb{E}[z_{t_\ell}^{(i)}]}{\text{std}(z_{t_\ell}^{(i)}) + \epsilon},$$

Finally, we filter low-support triples using minimum activation and co-occurrence thresholds, apply the pruning procedure described in Appendix Section A.2, annotate triples with residual-feature component labels, and write the ranked atlas to disk.

Stage three validates the triples using ablation. For each selected triple (s_k, u_j, t_ℓ) , we rerun the model and identify token positions where both the state feature s_k and the MLP update feature u_j are active. We then ablate only the decoded contribution of u_j from the MLP output, reconstruct the patched residual stream and re-encode it with the layer- i residual SAE. The normalized ablation score measures the ablation effect in units of the target feature’s own standard deviation:

$$A_z(s_k, u_j, t_\ell) = \frac{\mathbb{E}[z_\ell^{(i)} \mid s_k, u_j] - \mathbb{E}[\tilde{z}_\ell^{(i)} \mid s_k, u_j]}{\text{std}(z_\ell^{(i)}) + \epsilon}.$$

As a smaller complementary experiment, we ablate selected state features s_k for high-ranked (s_k, u_j) pairs and measure the normalized effect on the corresponding update feature u_j .

We normalize ablation effects because SAE feature activations are not naturally comparable across features. Some feature activations are much larger or more variable than others (Sun et al., 2024). Without normalization, high-scale target features would dominate the validation results even when their relative causal effect is small.

2.2 COSINE MATCHING

Decoder cosine similarity provides a natural geometric baseline because an SAE decoder vector represents the direction in activation space that a feature contributes. A high cosine similarity, therefore, indicates that two features write in similar directions, independent of their vector magnitudes. Moreover, cosine similarity can be computed directly from the SAE weights without collecting activation statistics or performing interventions. The issue with cosine matching, described by Balcells et al. (2024), is that feature-similarity measures cannot establish that an upstream feature causally contributes to a downstream feature. It measures whether two SAE features write in similar directions in activation space. It does not test whether the features are active on the same tokens, whether one feature actually changes the other, or whether the relation depends on a previous residual stream state. Thus, two individually low-cosine features can still jointly create a target feature when their combined contributions move the residual stream into the target feature’s activation region.

3 PRELIMINARY RESULTS

The 20M-token Pythia-160M $L_7 \rightarrow L_8$ run produced 262,735 filtered MLP transition triples. Of these, 38,125 have a strong ablation effect under the criterion $|A_z| \geq 2$, corresponding to 14.5% of the filtered triples. We use these validated triples to test whether strong feature transitions can be recovered from pairwise decoder geometry alone.

Our results show that high update-target cosine similarity captures direct MLP write relations. For $c_{u,t} \geq 0.7$, 2,254 of 3,211 triples reach $|A_z| \geq 2$ (70.2%). This trend remains visible at lower cosine thresholds. For $c_{u,t} \geq 0.5$, 3,876 of 5,855 triples reach $|A_z| \geq 2$ (66.2%). This confirms the findings by Laptev et al. (2025). Based on the high cosine similarity, we hypothesize that in triples with high update-target cosine similarity, a new feature is created by the update feature. The update-to-target results above measure whether ablating the update feature changes the target feature. We also run a smaller-sized state-to-update experiment, where the state feature is ablated, and the MLP update feature is measured. Across the 4,096 highest-ranked unique (s, u) pairs and 500k tokens, 3,765 pairs with at least ten active tokens reach $A_z \geq 2$ (91.9%), with a median update-feature effect of $A_z = 40.6$. This suggests that state features can strongly condition MLP update features, motivating the triple-level analysis below.

However, the pairwise decoder’s cosine similarity misses the strongest causal triples. Of the 38,125 triples with strong ablation effects, 33,550 have both $c_{s,t} < 0.7$ and $c_{u,t} < 0.7$, meaning that 88.0% of strong causal transitions would be missed by a high-cosine threshold on either the state-target or update-target pair. This remains true even when the threshold is substantially lowered. At $\tau = 0.4$, 23,677 of 38,125 strong-effect triples still have both pairwise cosine scores below threshold (62.1%). We use $\tau = 0.4$ and $\tau = 0.7$ as reference thresholds following Paulo & Belrose (2025). Figure 2 shows that the validated triples are not concentrated in the corresponding high-cosine regions. Instead, the dominant mass lies at a low state-target cosine, with $c_{s,t}$ close to zero, and only a moderate update-target cosine. Thus, for many validated transitions, neither the previous residual feature nor the MLP update feature is individually a high-cosine match to the target feature. The plot also shows a smaller high- $c_{u,t}$ band, which is consistent with direct write-like transitions where the update feature decoder is already aligned with the target decoder. However, this direct-write regime is only one subpopulation. Most strong causal transitions occupy lower-cosine regions, indicating that their relationship is better described as a state-conditioned interaction than as a pairwise geometric match. This supports the need for a transition atlas that scores triples $s_k + u_j \rightarrow t_\ell$, rather than relying only on pairwise decoder cosine similarity. Thus, in the full Pythia atlas, the answer to our research question is negative. Pairwise decoder cosine alone does not recover most strong causal feature transitions.

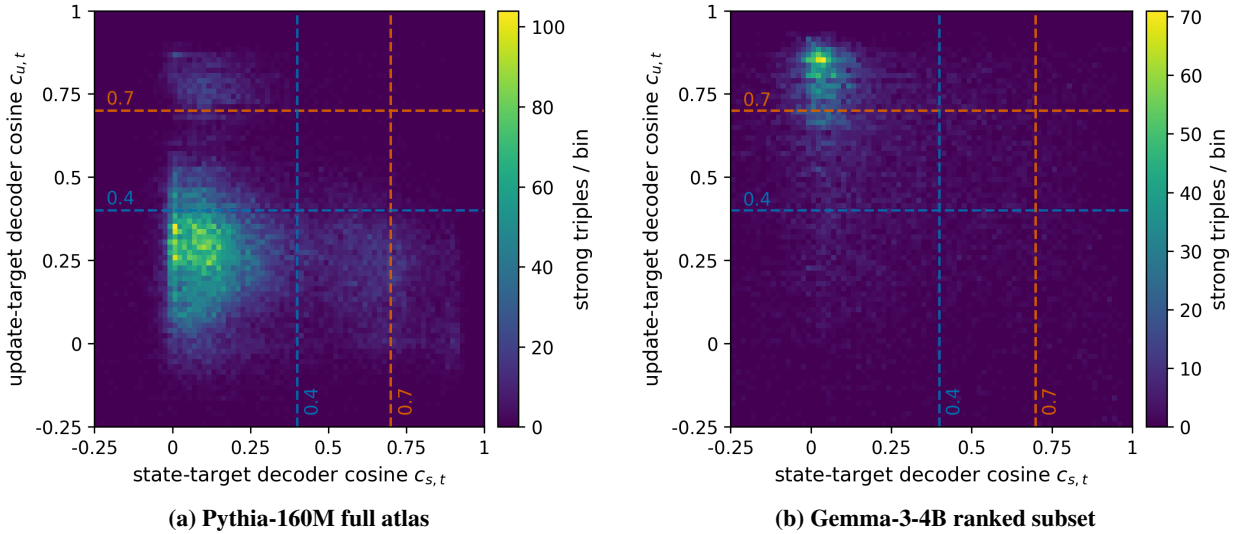


Figure 2: Strong causal ($|A_z| \geq 2$) transition triples in decoder-cosine space. Panel (a) shows the Pythia-160M $L_7 \rightarrow L_8$ full-atlas run. Panel (b) shows the Gemma-3-4B $L_{21} \rightarrow L_{22}$ validation subset, restricted to the top 30,000 ranked candidates. Dashed lines mark pairwise cosine thresholds $\tau = 0.4$ and $\tau = 0.7$. The Pythia panel shows that most strong transitions are missed by high pairwise-cosine thresholds; the Gemma panel provides a qualified cross-model check rather than a full-atlas recall measurement.

As a cross-model check, we evaluated Gemma-3-4B $L_{21} \rightarrow L_{22}$ using pretrained Gemma Scope residual and MLP SAEs (McDougall et al., 2025; Team et al., 2025). The atlas contains 231,198 filtered MLP triples, but to reduce runtime, we only validated the top 30,000 causal triples, so this is not a full-atlas validation. Restricting the analysis to strong-effect triples, there are 12,139 candidates with $|A_z| \geq 2$. Among these strong causal candidates, 6,506 (53.6%) have both $c_{s,t} < 0.7$ and $c_{u,t} < 0.7$, and 2,291 (18.9%) still have both pairwise cosine scores below 0.4. A top-30k update-target cosine baseline overlaps with 8,278 strong-effect candidates (68.2%) and misses 3,861 (31.8%), while the state-target baseline overlaps with only 1,676 (13.8%). Thus, the Gemma result is weaker than the Pythia result. It still shows that many strong triples are not recovered by state-target cosine, but update-target cosine explains a much larger fraction of the strongest effects, and the subset validation prevents a full-atlas recall claim.

4 LIMITATIONS AND ONGOING WORK

Our strongest evidence is the completed $L_7 \rightarrow L_8$ Pythia-160M full-atlas run using custom-trained SAEs. The Gemma-3-4B result provides only qualified cross-model support, since it causally validates only the top 30,000 ranked triples, and its strongest effects are more often recovered by update-target cosine. It therefore remains open how much the observed low recall of decoder cosine similarity depends on model scale, layer choice, SAE architecture, pruning parameters, and validation budget.

We use these limitations to set the priorities for our future work. As we currently only have estimates of how different pruning parameters affect the atlas, we plan to compare additional scoring functions for ranking triples before causal validation. We also aim to validate more Gemma Scope (Team et al., 2025) layer transitions to test whether context-conditioned feature-flow relations appear consistently across layers. Besides improving scalability, the usage of pretrained SAEs could also enable the adoption of tools like Neuronpedia (Lin, 2023) to gain deeper insight into SAE features.

Creating a feature-flow atlas for one or more LLMs would open up many opportunities to generate new hypotheses about how individual features work. Although this study focuses on inference dynamics within a fixed model, we believe feature-flow analysis could also become a useful tool for studying learning dynamics in continually adapted foundation models. Existing continual-learning approaches mainly analyze parameter updates or representation-level changes, whereas feature-flow gives access to the computational interactions that drive representation evolution. Comparing feature-flow atlases across successive stages of adaptation may therefore offer a complementary mechanistic perspective on transfer, interference, and knowledge integration in foundation models.

REFERENCES

- Daniel Balcells, Benjamin Lerner, Michael Oesterle, Ediz Ucar, and Stefan Heimersheim. Evolution of sae features across layers in llms, 2024. URL <https://arxiv.org/abs/2410.08869>.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling, 2023. URL <https://arxiv.org/abs/2304.01373>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023. URL <https://arxiv.org/abs/2309.08600>.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017.
- Daniil Laptev, Nikita Balagansky, Yaroslav Aksenov, and Daniil Gavrilov. Analyze feature flow to enhance interpretation and steering in language models, 2025. URL <https://arxiv.org/abs/2502.03032>.
- Johnny Lin. Neuronpedia: Interactive reference and tooling for analyzing neural networks, 2023. URL <https://www.neuronpedia.org>. Software available from neuronpedia.org.
- Callum McDougall, Arthur Conmy, János Kramár, Tom Lieberum, Senthooan Rajamanoharan, and Neel Nanda. Gemma scope 2: Technical paper. *Google DeepMind Blog*, 2025.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt, 2023. URL <https://arxiv.org/abs/2202.05262>.
- Julian Minder, Clément Dumas, Caden Juang, Bilal Chughtai, and Neel Nanda. Overcoming sparsity artifacts in crosscoders to interpret chat-tuning, 2025. URL <https://arxiv.org/abs/2504.02922>.
- Gonçalo Paulo and Nora Belrose. Sparse autoencoders trained on the same data learn different features, 2025. URL <https://arxiv.org/abs/2501.16615>.
- Xiangchen Song, Aashiq Muhamed, Yujia Zheng, Lingjing Kong, Zeyu Tang, Mona T. Diab, Virginia Smith, and Kun Zhang. Position: Mechanistic interpretability should prioritize feature consistency in saes, 2025. URL <https://arxiv.org/abs/2505.20254>.
- Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. Massive activations in large language models, 2024. URL <https://arxiv.org/abs/2402.17762>.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, Thomas Mesnard, Geoffrey Cideron, Jean bastien Grill, Sabela Ramos, Edouard Yvinec, Michelle Casbon, Etienne Pot, Ivo Penchev, Gaël Liu, Francesco Visin, Kathleen Kenealy, Lucas Beyer, Xiaohai Zhai, Anton Tsitsulin, Robert Busa-Fekete, Alex Feng, Naveen Sachdeva, Benjamin Coleman, Yi Gao, Basil Mustafa, Iain Barr, Emilio Parisotto, David Tian, Matan Eyal, Colin Cherry, Jan-Thorsten Peter, Danila Sinopalnikov, Surya Bhupatiraju, Rishabh Agarwal, Mehran Kazemi, Dan Malkin, Ravin Kumar, David Vilar, Idan Brusilovsky, Jiaming Luo, Andreas Steiner, Abe Friesen, Abhan-shu Sharma, Abheesht Sharma, Adi Mayrav Gilady, Adrian Goedeckemeyer, Alaa Saade, Alex Feng, Alexander Kolesnikov, Alexei Bendebury, Alvin Abdagic, Amit Vadi, András György, André Susano Pinto, Anil Das, Ankur Bapna, Antoine Miech, Antoine Yang, Antonia Paterson, Ashish Shenoy, Ayan Chakrabarti, Bilal Piot, Bo Wu, Bobak Shahriari, Bryce Pettrini, Charlie Chen, Charline Le Lan, Christopher A. Choquette-Choo, CJ Carey, Cormac Brick, Daniel Deutsch, Danielle Eisenbud, Dee Cattle, Derek Cheng, Dimitris Paparas, Divyashree Shivakumar Sreepathihalli, Doug Reid, Dustin Tran, Dustin Zelle, Eric Noland, Erwin Huizenga, Eugene Kharitonov, Frederick Liu, Gagik Amirkhanyan, Glenn Cameron, Hadi Hashemi, Hanna Klimczak-Plucińska, Harman Singh, Harsh Mehta, Harshal Tushar Lehri, Hussein Hazimeh, Ian Ballantyne, Idan Szpektor, Ivan Nardini, Jean Pouget-Abadie, Jetha Chan, Joe Stanton, John Wieting, Jonathan Lai, Jordi Orbay, Joseph Fernandez, Josh Newlan, Ju yeong Ji, Jyotinder Singh, Kat Black, Kathy Yu, Kevin Hui, Kiran Vodrahalli, Klaus Greff, Linhai Qiu, Marcella Valentine, Marina Coelho, Marvin Ritter, Matt Hoffman, Matthew Watson, Mayank Chaturvedi, Michael Moynihan, Min Ma, Nabila Babar, Natasha Noy, Nathan Byrd, Nick Roy, Nikola Momchev, Nilay Chauhan, Naveen Sachdeva, Oskar Bunyan, Pankil Botarda, Paul Caron, Paul Kishan Rubenstein, Phil Culliton, Philipp Schmid, Pier Giuseppe Sessa, Pingmei Xu, Piotr Stanczyk, Pouya Tafti, Rakesh Shivanna, Renjie Wu, Renke Pan, Reza

Rokni, Rob Willoughby, Rohith Vallu, Ryan Mullins, Sammy Jerome, Sara Smoot, Sertan Girgin, Shariq Iqbal, Shashir Reddy, Shruti Sheth, Siim Pöder, Sijal Bhatnagar, Sindhu Raghuram Panyam, Sivan Eiger, Susan Zhang, Tianqi Liu, Trevor Yacovone, Tyler Liechty, Uday Kalra, Utku Evci, Vedant Misra, Vincent Roseberry, Vlad Feinberg, Vlad Kolesnikov, Woohyun Han, Woosuk Kwon, Xi Chen, Yinlam Chow, Yuvein Zhu, Zichuan Wei, Zoltan Egyed, Victor Cotruta, Minh Giang, Phoebe Kirk, Anand Rao, Kat Black, Nabila Babar, Jessica Lo, Erica Moreira, Luiz Gustavo Martins, Omar Sanseviero, Lucas Gonzalez, Zach Gleicher, Tris Warkentin, Vahab Mirrokni, Evan Senter, Eli Collins, Joelle Barral, Zoubin Ghahramani, Raia Hadsell, Yossi Matias, D. Sculley, Slav Petrov, Noah Fiedel, Noam Shazeer, Oriol Vinyals, Jeff Dean, Demis Hassabis, Koray Kavukcuoglu, Clement Farabet, Elena Buchatskaya, Jean-Baptiste Alayrac, Rohan Anil, Dmitry, Lepikhin, Sebastian Borgeaud, Olivier Bachem, Armand Joulin, Alek Andreev, Cassidy Hardin, Robert Dadashi, and Léonard Hussenot. Gemma 3 technical report, 2025. URL <https://arxiv.org/abs/2503.19786>.

Buddhi Wickramasinghe, Gobinda Saha, and Kaushik Roy. Continual learning: A review of techniques, challenges, and future directions. *IEEE Transactions on Artificial Intelligence*, 5(6):2526–2546, 2023.

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Zican Dong, Yupeng Hou, Beichen Zhang, Yingqian Min, Junjie Zhang, Peiyu Liu, et al. A survey of large language models. *Frontiers of Computer Science*, 20(12):2012627, 2026.

Junhao Zheng, Shengjie Qiu, Chengming Shi, and Qianli Ma. Towards lifelong learning of large language models: A survey. *ACM Computing Surveys*, 57(8):1–35, 2025.

A APPENDIX

A.1 NOTATION AND SCORE DEFINITIONS

A.1.1 SYMBOLS

Symbol	Meaning
i	Target transformer layer.
$r_{\text{post}}^{(i-1)}$	Residual stream before the target layer.
$r_{\text{post}}^{(i)}$	Residual stream after the target layer.
$a^{(i)}$	Attention output of layer i .
$m^{(i)}$	MLP output of layer i .
$\tilde{m}^{(i)}$	MLP output after removing the decoded update feature.
$E_{\text{res}}^{(i)}$	Residual-stream SAE encoder for layer i .
$E_{\text{mlp}}^{(i)}$	MLP-output SAE encoder for layer i .
s_k	Layer- $(i-1)$ residual state feature.
u_j	Layer- i MLP update feature.
t_ℓ	Layer- i residual target feature.
$z_\ell^{(i)}$	Activation of target feature t_ℓ after layer i .
$\tilde{z}_\ell^{(i)}$	Target-feature activation after patching.
$\Delta z_\ell^{(i, \text{mlp})}$	MLP-induced change in target feature t_ℓ .
$d_{s_k}, d_{u_j}, d_{t_\ell}$	Decoder directions for the state, update, and target features.
A_z	Normalized ablation effect.
$c_{s,t}, c_{u,t}$	State-target and update-target decoder-cosine baselines.
ϵ	Small constant for numerical stability.
τ	Pairwise decoder-cosine threshold, e.g. 0.4 or 0.7.

A.1.2 LAYER DECOMPOSITION

For each token, the residual stream after layer i is approximated by

$$r_{\text{post}}^{(i)} \approx r_{\text{post}}^{(i-1)} + a^{(i)} + m^{(i)}.$$

A.1.3 MLP-INDUCED TARGET CHANGE

We encode both the original residual stream and the no-MLP residual stream in the same layer- i residual SAE basis:

$$z_{\text{after}}^{(i)} = E_{\text{res}}^{(i)}(r_{\text{post}}^{(i)}), \quad z_{\text{no-mlp}}^{(i)} = E_{\text{res}}^{(i)}(r_{\text{post}}^{(i)} - m^{(i)}).$$

The MLP-induced target-feature change is

$$\Delta z^{(i, \text{mlp})} = z_{\text{after}}^{(i)} - z_{\text{no-mlp}}^{(i)}.$$

A.1.4 DECODER-COSINE BASELINES

Let d_{s_k} , d_{u_j} , and d_{t_ℓ} be decoder directions for the state, update, and target features. The cosine baselines are

$$c_{s,t} = \frac{d_{s_k}^\top d_{t_\ell}}{\|d_{s_k}\|_2 \|d_{t_\ell}\|_2}, \quad c_{u,t} = \frac{d_{u_j}^\top d_{t_\ell}}{\|d_{u_j}\|_2 \|d_{t_\ell}\|_2}.$$

A.1.5 CAUSAL VALIDATION

For a selected triple (s_k, u_j, t_ℓ) , we validate the candidate transition token positions where both s_k and u_j are active. We remove the decoded update-feature contribution from the MLP output,

$$z_{u_j}^{(i, \text{mlp})} d_{u_j}^{(i, \text{mlp})},$$

and reconstruct the patched residual stream

$$\tilde{r}_{\text{post}}^{(i)} = r_{\text{post}}^{(i-1)} + a^{(i)} + \tilde{m}^{(i)}.$$

Let $\tilde{z}_\ell^{(i)}$ be the target-feature activation after patching. The normalized update-to-target ablation effect is

$$A_z(s_k, u_j, t_\ell) = \frac{\mathbb{E}[z_\ell^{(i)} | s_k, u_j] - \mathbb{E}[\tilde{z}_\ell^{(i)} | s_k, u_j]}{\text{std}(z_\ell^{(i)}) + \epsilon}.$$

A triple is counted as *validated* when its intervention produces a large normalized target-feature change, using the magnitude-only criterion $|A_z| \geq 2$. This is the validation criterion used throughout the main analysis, because the question there is whether pairwise decoder geometry recovers transitions with large intervention effects. We do not require the atlas score to predict the sign of the ablation effect.

A.1.6 STATE-TO-UPDATE CHECK

As a complementary check, we also test whether state features causally condition MLP update features. For selected (s_k, u_j) pairs, we ablate the decoded state-feature contribution and measure the normalized change in the corresponding MLP SAE update feature u_j . This check validates the causal relation $s_k \rightarrow u_j$, showing that the MLP update feature can depend on the prior residual-state feature.

A.2 PRUNING

A full enumeration of all possible transition triples is computationally infeasible. In principle, each token could activate many SAE features with different strengths, so the search space per token scales with the product of the layer- $(i-1)$ residual SAE dimension, the layer- i MLP-out SAE dimension, and the layer- i residual SAE dimension. We therefore restrict the token-level search space using the number of layer- $(i-1)$ state features, layer- i update features, and layer- i target features retained per token as parameters. For the Pythia run, we retain the top 64 active features per token during event extraction, and for Gemma we validate only the top 30,000 ranked causal triples. For each token, we keep the top active state and update features, and a larger set of candidate target features. The raw candidate set is then the Cartesian product of these sparse feature sets. We further prune this set by retaining only the strongest n local candidate triples per token, ranked by

$$|z_{s_k} \cdot z_{u_j} \cdot \Delta z_\ell^{(i, \text{mlp})}|,$$

where z_{s_k} is the layer- $(i-1)$ state activation, z_{u_j} is the layer- i MLP update activation, and $\Delta z_\ell^{(i, \text{mlp})}$ is the MLP-induced change in the layer- i target feature.

This token-level pruning is separate from the later validation budget, which determines how many ranked triples are interventionally tested. With this pruning, the candidate atlas should be understood as a precision-oriented discovery atlas rather than an exhaustive enumeration of all possible feature transitions.

Quantity	Pythia-160M $L_7 \rightarrow L_8$	Gemma-3-4B $L_{21} \rightarrow L_{22}$
Atlas/scoring tokens	20M	20M
Unfiltered MLP candidate triples	199,004,232	146,283,098
Filtered MLP triples	262,735	231,198
Validation tokens	4M	4M
Validated interaction-ranked triples	262,735	30,000
Strong ablation-effect triples ($ A_z \geq 2$)	38,125	12,139

Table 1: Pipeline and validation summary for the Pythia-160M full-atlas run and the Gemma-3-4B ranked validation subset.

A.3 PYTHIA SAE TRAINING HYPERPARAMETERS

The table lists the training configuration for the custom Pythia-160M SAEs. For Gemma, we use pretrained width-16k, L_0 -big Gemma Scope 2 residual SAEs at layers 21 and 22 and the layer-22 MLP SAE.

Hyperparameter	Value
Model	EleutherAI/pythia-160m
Dataset	monology/pile-uncopyrighted
SAE architecture	TopK SAE
Dictionary size d_{SAE}	33024
Input dimension d_{model}	768
TopK sparsity K_{top}	100
Training tokens	8B
Context length	128
Token batch size	4096
Learning rate	$3 \cdot 10^{-4}$
Adam betas	(0.9, 0.999)
Learning-rate warmup	500 steps
Auxiliary loss coefficient	1.0
Training seed	43

Table 2: Training hyperparameters for the custom Pythia-160M SAEs.